

NDNComm 2021, NIST (Virtual)
October 28-29

BLEnD: Improving NDN Performance Over Wireless Links Using Interest Bundling

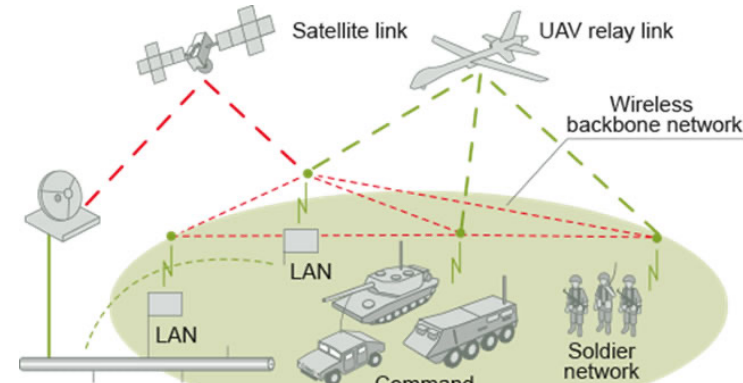
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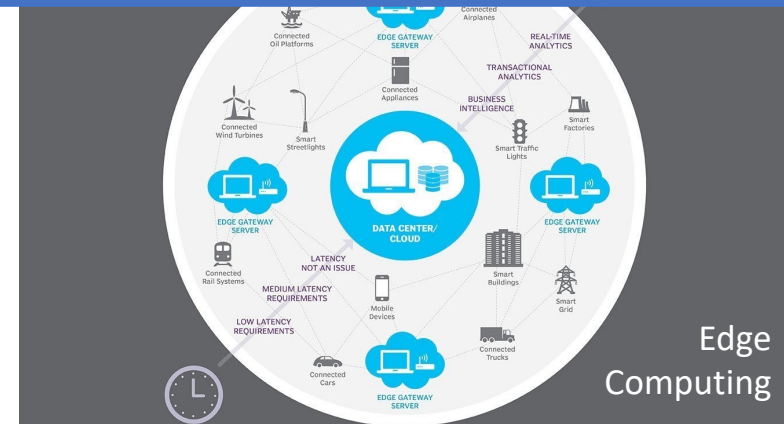
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Dawn of the Planet of the Wireless



Expectation: High Throughput and Reliability



Wireless Communication

- Half-duplex → Either send or receive.
- Channel access contention → Pause send until channel available.
- Other challenges
 - Hidden terminals → collision.
 - Congestion.

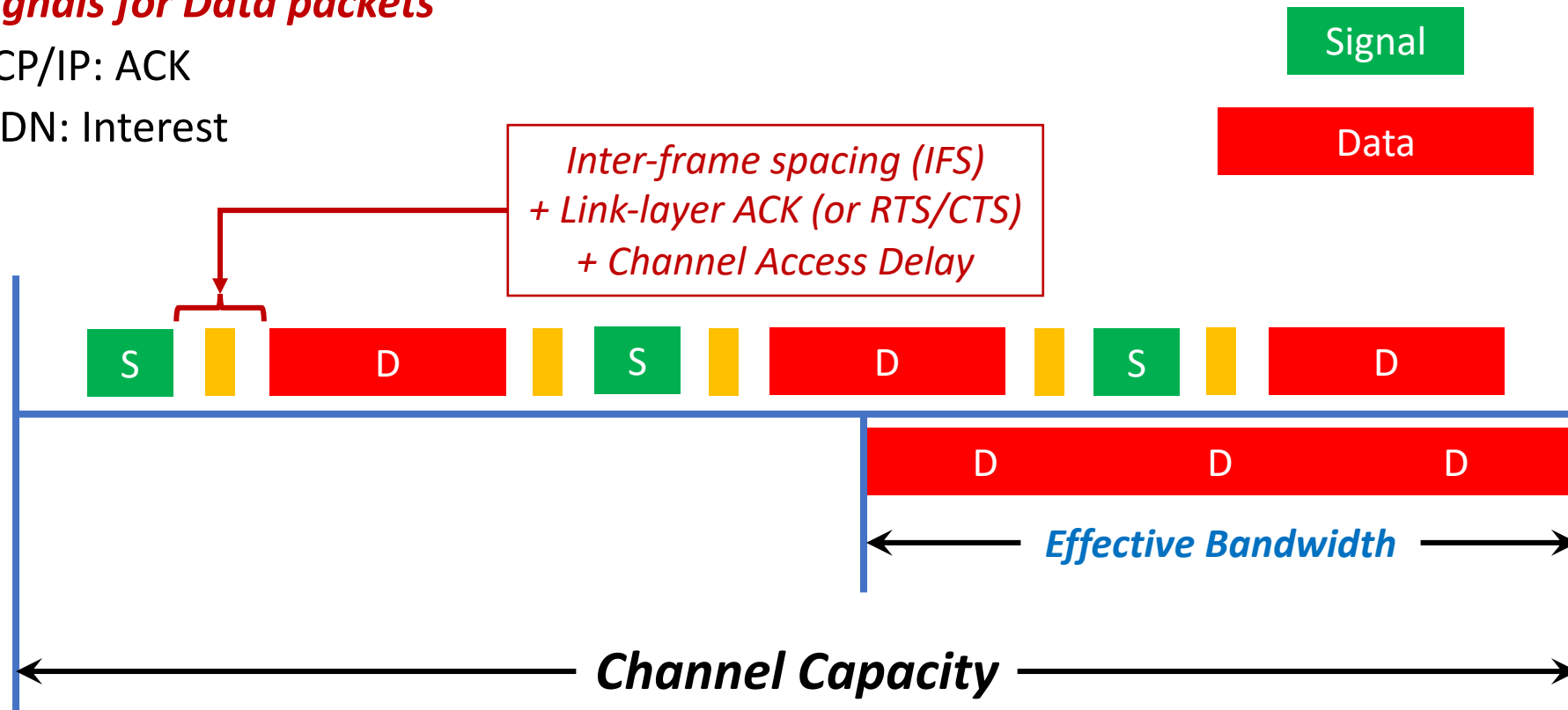
Data Exchange in Wireless Medium

- Congestion control, loss recovery

- *Signals for Data packets*

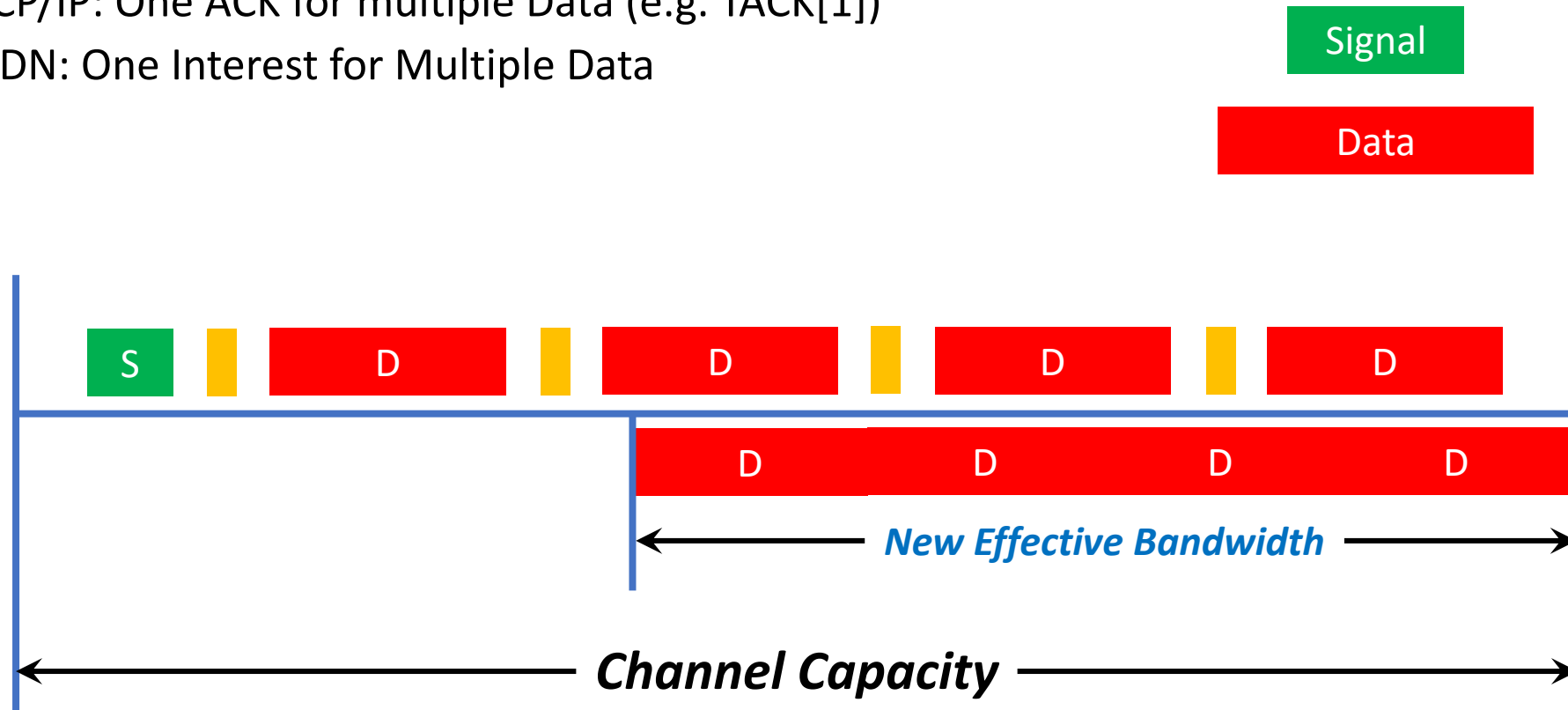
- TCP/IP: ACK

- NDN: Interest



Increasing Channel Access for Data

- Reduce *signal* frequency
 - TCP/IP: One ACK for multiple Data (e.g. TACK[1])
 - NDN: One Interest for Multiple Data

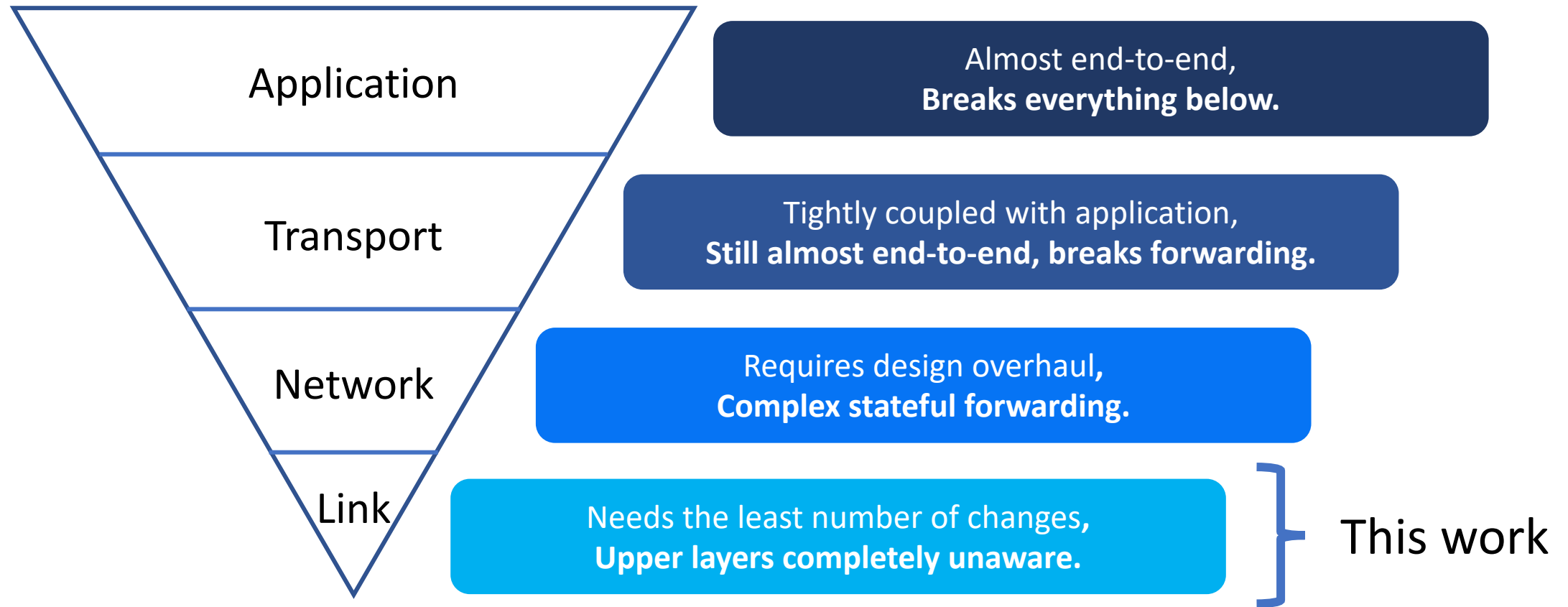


Interest Bundling Improves Throughput

- Single-hop Wi-Fi
 - Infrastructure mode, AP is the producer.
- Testing with custom application-
 - Single Interest fetches continuous data.

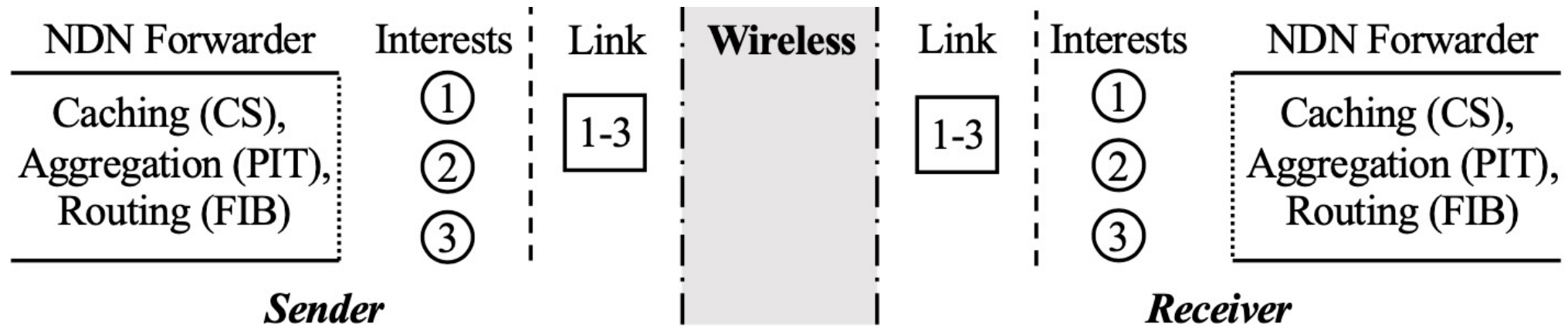
	802.11b at 11 Mbps	802.11n at 24 Mbps
Default or no bundle (Mbps)	4.4	9.7
One Interest (Mbps)	7.0	18.6
Goodput Improvement	59.1%	91.7%

...But where to Bundle/Unbundle?



BLEnD: High-level Idea

Bundling Interests with **L**ink-layer **En**coding and **D**ecoding

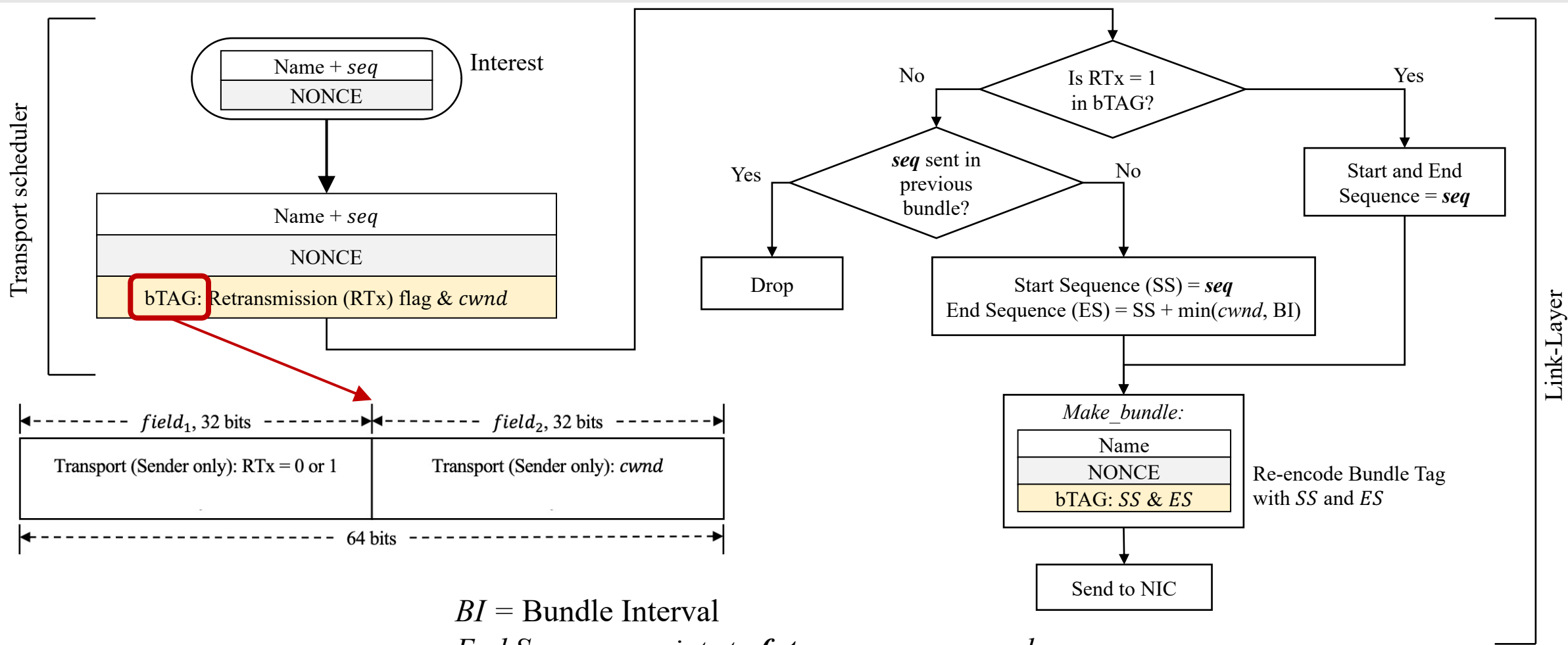




BLEnD: Encoding and Decoding

- Sender-side encoding
- Receiver-side decoding

Sender-side Encoding

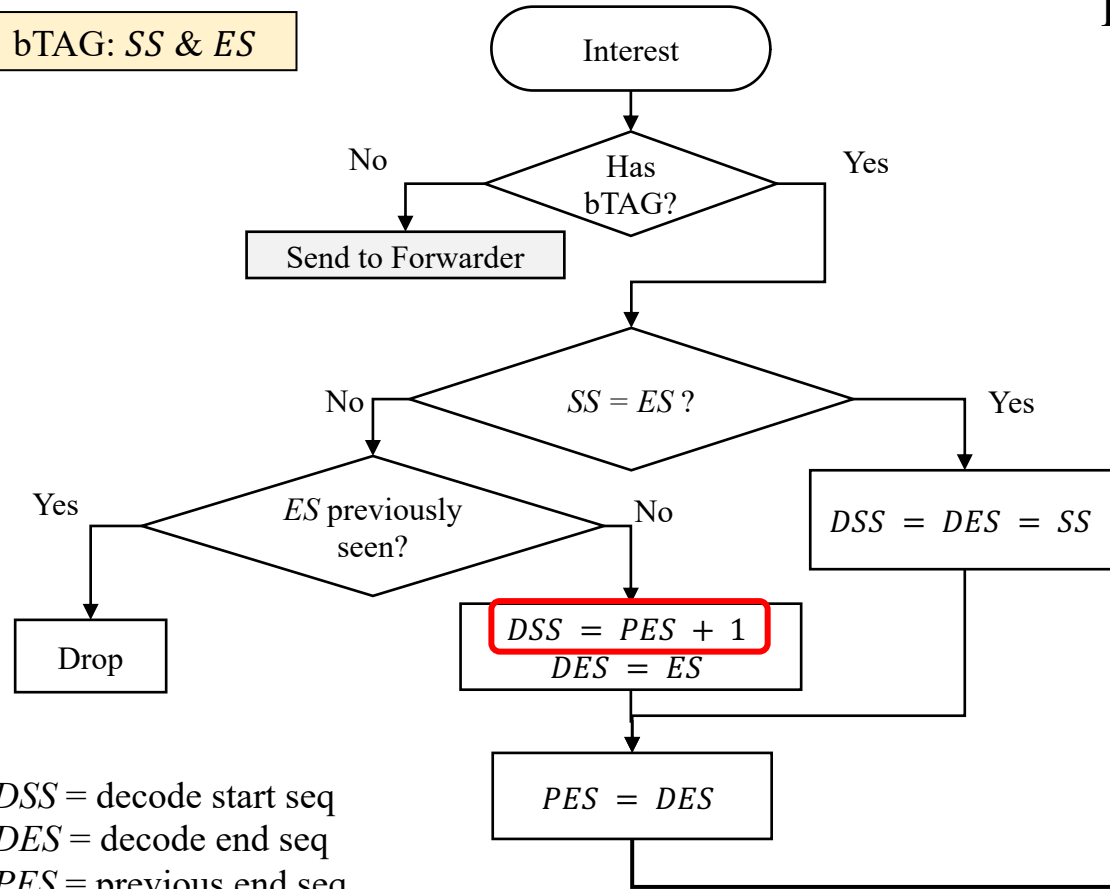


BI = Bundle Interval

End Sequence points to **future** sequence numbers

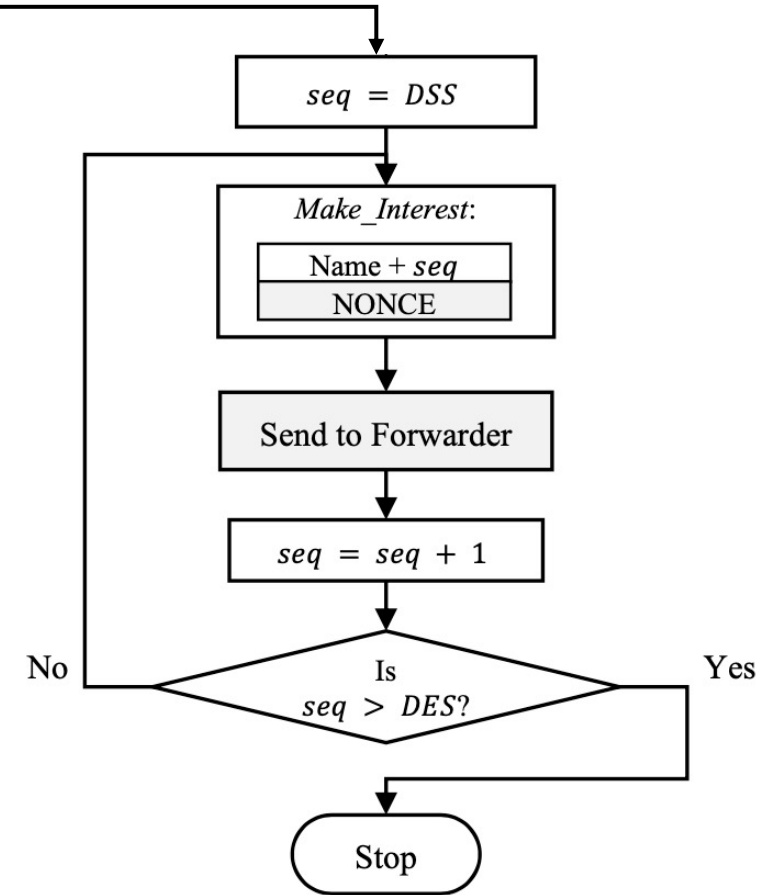
Receiver-side Decoding

bTAG: *SS* & *ES*



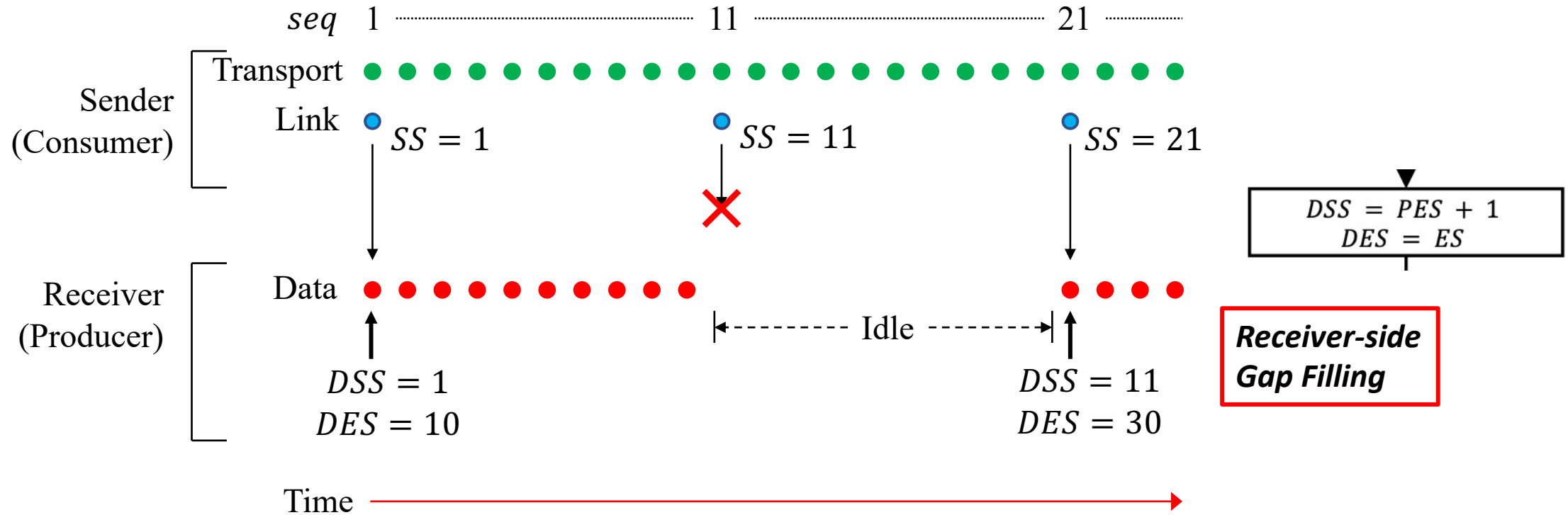
DSS = decode start seq
DES = decode end seq
PES = previous end seq

Link-layer



Handling Lost Bundled Interest

**** Bundle Interval (BI) = 10 at the sender**





Simulation Results

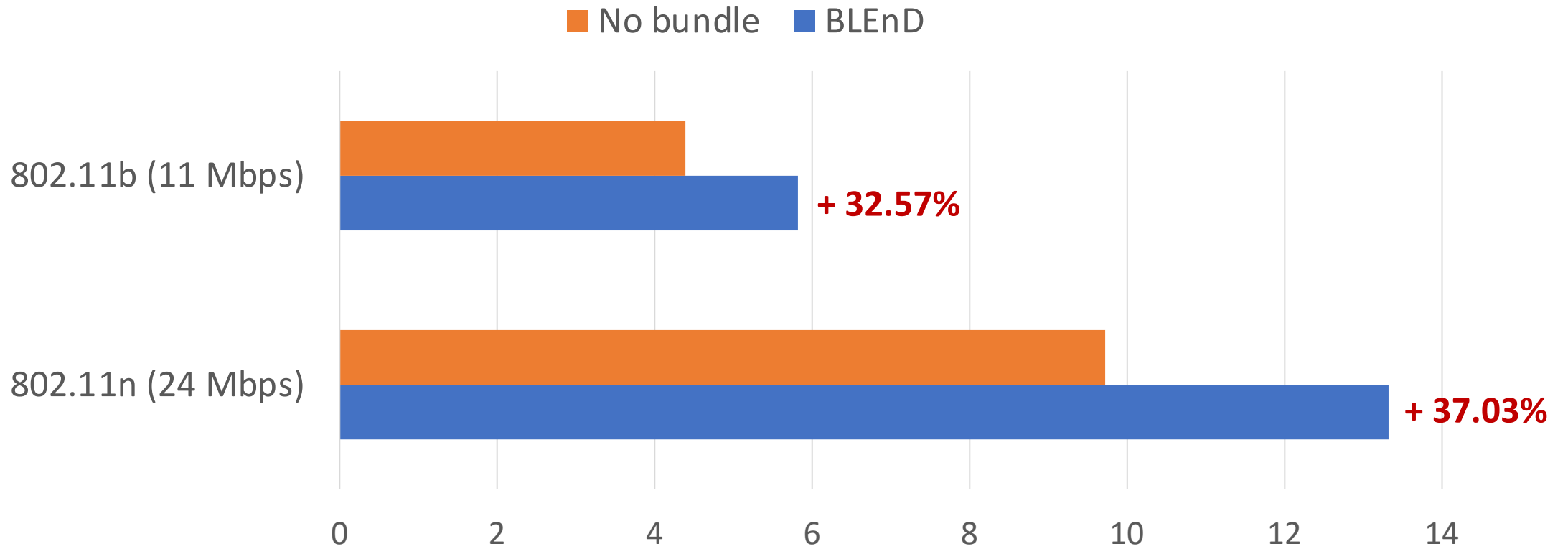
- Goodput Improvement
- Effect of Bundle Interval (or Size)

ndnSIM

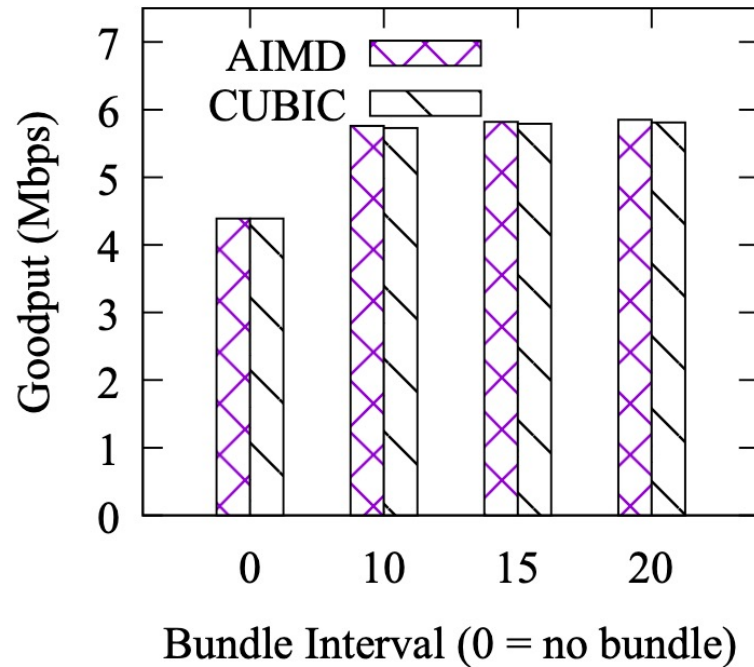
Single-hop (Station and Access-point)

BLEnD- *Face::GenericLinkService*
bTAG- *Lp::tag*

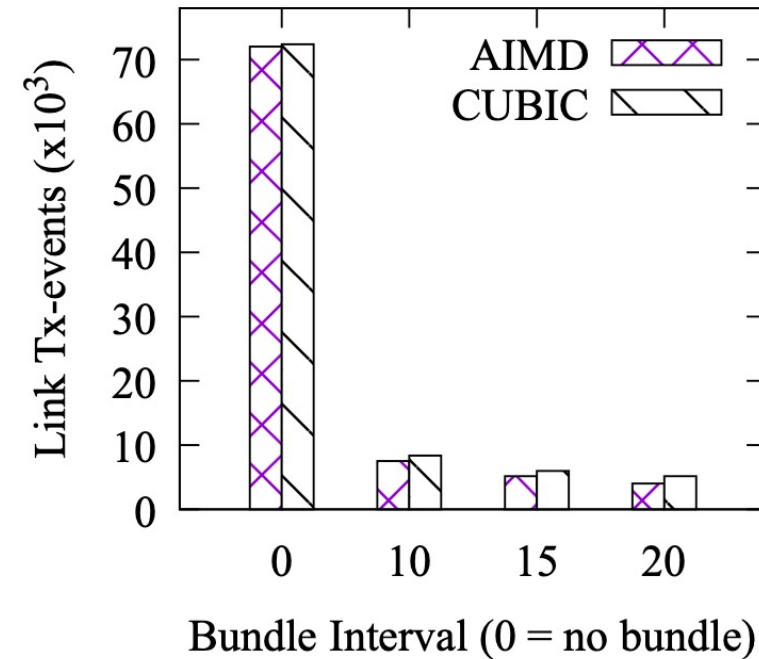
Goodput Improvement



Effect of Bundle Interval (or Size)



(a) Goodput.



(b) Link-layer Tx-events.

*Using 802.11b (11 Mbps). Bundle Interval (BI) = 0 represents no Interest bundling.
Sender's Link-layer transmissions (Tx) include application or transport retransmission packets*



Some Challenges

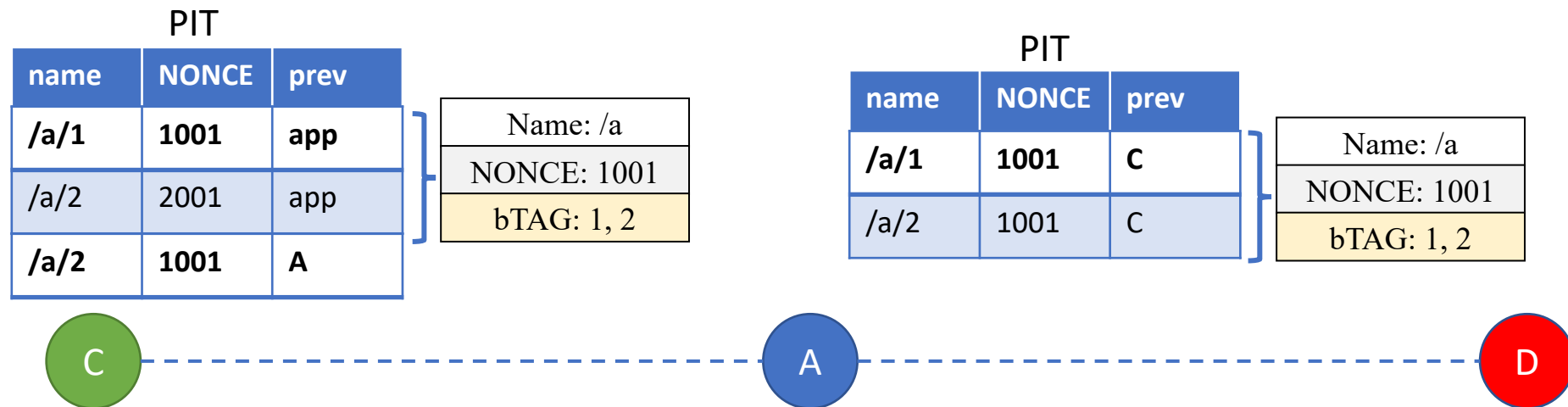
- RTT fluctuation
- Duplicate data in multi-hop

RTT Fluctuation

- Producer processes bundled Interest immediately and sends data.
- Receiver-side lost bundled-Interest recovery is a delayed reaction.
- Proof: Changing RTT variance multiplier (γ) and using 802.11n (Default $\gamma = 4$)
 - File = 100MB, Bundle Size = 15,
 - Expected Data packets = 71821 (each chunk payload = 1460bytes)

	$\gamma =$	RTx	App_{sent}	Goodput (Mbps)
AIMD	4	629	72450	13.32
	10	590	72411	13.33
	20	580	72401	13.34
CUBIC	4	2567	74388	13.1
	10	1982	73803	13.2
	20	1616	73437	13.25

Duplicate Data in Multi-hop



On Data, **C** will also forwards to **A**

- Keep a **<Name, NONCE>** table in sender's link-layer
 - Keep track of sent bundles' NONCEs.
 - Happens at discovery phase, table size should be small (low overhead).

More Research Opportunities

Rate-based congestion controller

- To avoid RTT fluctuation.

Dynamic bundle-interval

- To scale well over different link capacity.

Sender-side bundle loss reaction

- Reduce producer idle time.

Reducing hidden-station effect

- Minimize loss.

Hop-by-hop transport

- For multi-hop networks (plus mobility).
 - Multi-path for load-balancing.
- Extend Sync for transport functionalities at each hop.
 - e.g., rate management.



Thank you!

Q&A

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References:

1. Tong Li et. al., “TACK: Improving wireless transport performance by taming acknowledgments,” SIGCOMM '20. New York, NY, USA.
2. Rahman et. al., “BLEnD: Improving NDN performance over wireless links using interest bundling,” MILCOM '21. San Diego, CA, USA.